

Marks: 70

[Answer any five questions. Assume reasonable value for any missing data.]

Time: 3.00 Hours

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| 2. a) What is diode? Implement NOR gate by diode. | 5 |
| b) Show that Transistor acts as a switch. | 4 |
| c) Describe Exclusive NOR gate by transistor. | 5 |
| 3. a) What do mean by LED and LCD. | 4 |
| b) Explain Sample and Hold (S/H) circuit by using Operational Amplifier. | 5 |
| c) Describe Analog to digital Converter circuit. | 5 |
| 4. a) What is PLA? Explain block diagram of PLA circuit. | 4 |
| b) Describe the pulse transformer circuit and its equivalent circuit. | 5 |
| c) Explain pulse transmission with input and output waveforms in different media. | 5 |
| 5. a) What do you mean by flip flops? Explain operating principle of J-K flip flop. | 4 |
| b) Draw & explain BCD ripple counter. | 5 |
| c) What is the difference between a serial and parallel transfer? | 2 |
| d) Explain operating principle of 3-bit register. | 3 |
| 6. a) What do you mean by A/D & D/A converters? | 2 |
| b) Draw the Transistor Transistor Logic (TTL) circuit and explain working procedure. | 5 |
| c) Explain operating principle of comparator circuit. | 4 |
| d) Write down a comparison between A/D & D/A converters. | 3 |
| 7. a) What do you mean by clippers & clampers circuit? | 2 |
| b) Describe Negative clipping circuit with output waveforms. | 5 |
| c) What is Schmitt trigger circuit? Explain operating principle of Schmitt trigger circuit. | 5 |
| d) Write down application of clippers circuit. | 2 |
| 8. a) What do you mean by multivibrators? | 2 |
| b) Explain operating principle of astable multivibrators. | 6 |
| c) Write down application of monostable multivibrators. | 2 |
| d) Briefly explain various methods of generating a time-base waveform. | 4 |